Customer Behavior Analysis – Task 1

Prepared for: Alfido Tech | **Role:** Data Analytics Intern | **Date:** October 2026

1. Project Objective & Dataset Overview

The objective of this analysis is to evaluate customer transaction behavior, establish meaningful behavioral segments using RFM (Recency, Frequency, Monetary) modeling, identify purchase trends and churn risks, and deliver five data-grounded recommendations for Alfido Tech.

Dataset Details: The analysis utilized an e-commerce customer transaction dataset covering **250,000 transaction records** across **49,673 unique customers** spanning January 1, 2020 through September 15, 2023. Key attributes include transaction dates, product categories (Books, Clothing, Electronics, Home), product prices, order quantities, total purchase amounts, payment channels, returns, customer demographics (age, gender), and churn labels.

2. Data Cleaning & Feature Engineering Summary

Step / Field	Issue Identified	Action Taken
Duplicate Check	0 exact duplicates found in 250k rows	Retained full dataset integrity
Returns Column	47,596 missing (NaN) values	Imputed with 0.0 (no return recorded)
Purchase Date	Stored as string format	Parsed to datetime for temporal features
Age Columns	Redundant 'Age' and 'Customer Age'	Removed duplicate 'Age' column
Customer RFM	Transaction-level granularity	Aggregated to customer level (Recency, Frequency, Monetary)

3. Customer Segmentation (RFM Methodology)

Using a reference date set to September 16, 2023 (one day past the latest transaction), each customer was scored on **Recency** (days since last purchase), **Frequency** (total lifetime orders), and **Monetary** value (total spend). Scores from 1 to 5 were assigned via quintiles (`pd.qcut`, with first-rank frequency ranking). Customers were then mapped into five distinct, intuitive behavioral segments.

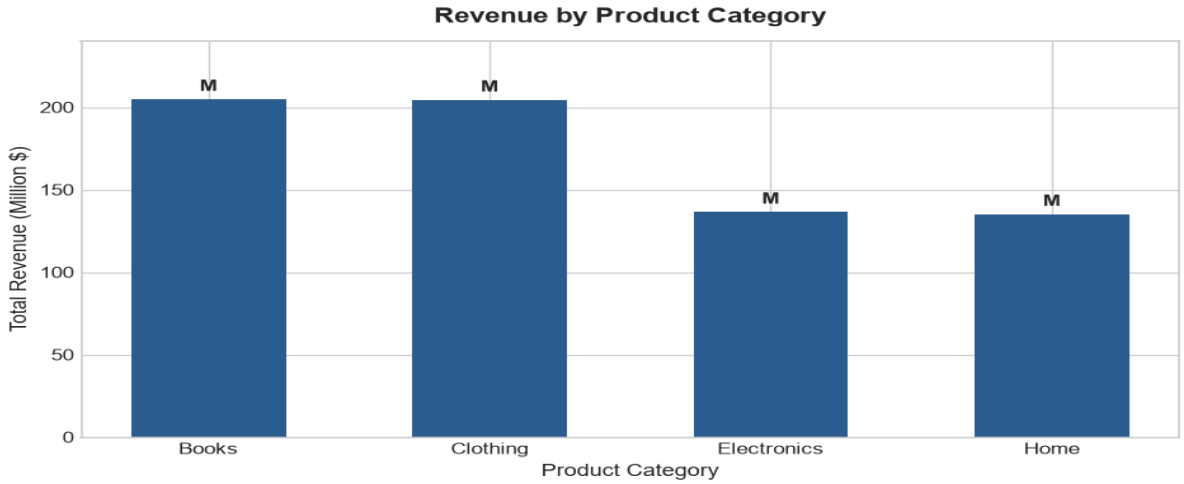
Segment Name	Customers	% Base	Avg Spend	Total Revenue	Avg Recency	Orders
Champions	8,739	17.6%	$21,785	$190.4M (27.9%)	61.3 days	7.65
Loyal Customers	12,354	24.9%	$15,355	$189.7M (27.8%)	129.0 days	5.87
At Risk	8,711	17.5%	$16,152	$140.7M (20.7%)	411.5 days	5.94
Lost / Hibernating	11,115	22.4%	$7,579	$84.2M (12.4%)	564.3 days	2.78
Potential / Recent	8,754	17.6%	$8,719	$76.3M (11.2%)	114.2 days	3.20

4. Key Findings: Purchase Patterns & Retention Trends

Monthly Sales & Repeat Purchase Health: Transaction volume remained consistently healthy between $14.5M and $15.6M per month across 2020 through mid-2023. Repeat purchase rate reached 96.61%, with 47,988 customers ordering multiple times.

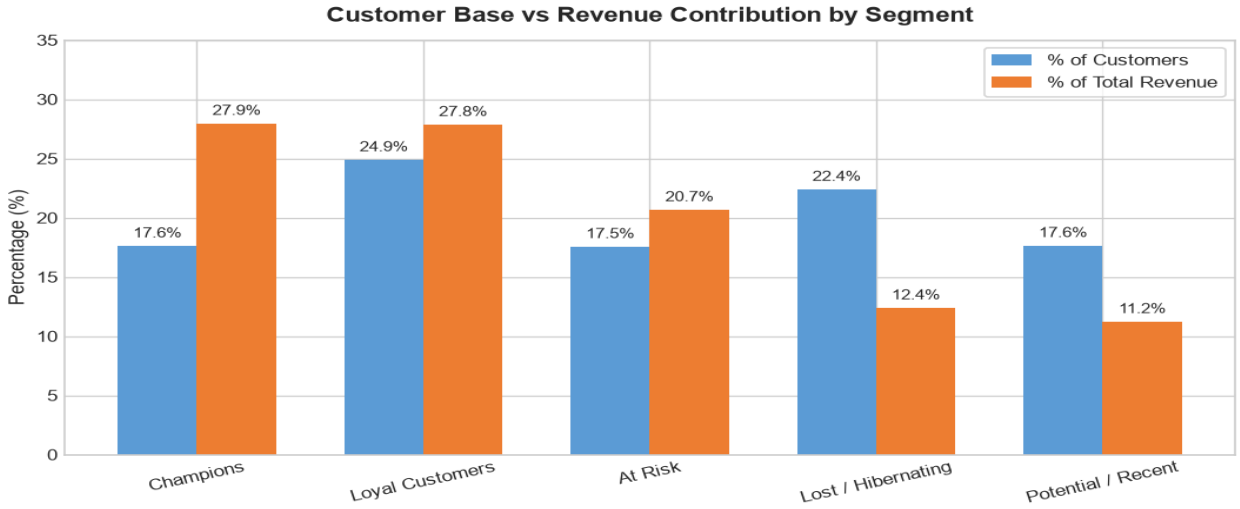


Category Demand & Revenue Concentration: Books ($204.9M, 74,912 orders) and Clothing ($204.5M, 75,052 orders) account for 60.0% of total revenue. However, returns average 49.8% across recorded orders, signaling an operational priority.

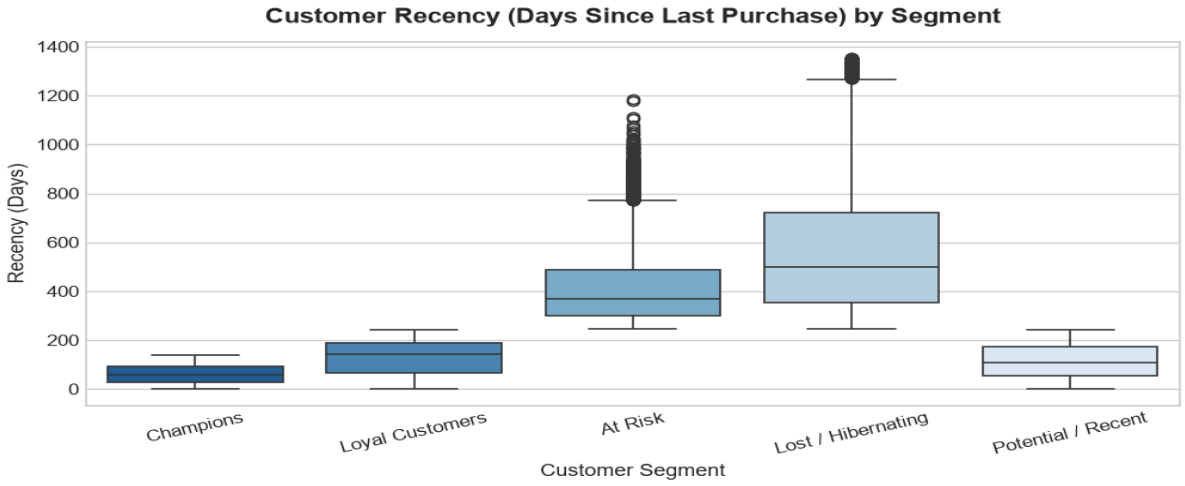


5. Segment Performance & Churn Risk Analysis

Revenue Skew: The top two segments (Champions and Loyal Customers) represent 42.46% of the customer base but drive 55.78% ($380.1M) of total revenue. In contrast, 8,711 At-Risk customers account for $140.7M in historical revenue but have lapsed for an average of 411.5 days.



Inactivity & Churn Dynamics: Overall customer churn is 20.01% (9,942 churned accounts). The distribution below shows that At-Risk and Lost customers demonstrate severe dormancy (recency > 400 days), requiring automated lifecycle intervention before 180 days.



6. Five Actionable Business Recommendations for Alfido Tech

Based on empirical findings across the 250,000 transactions, the following five targeted actions are recommended:

Recommendation 1 – Targeted Win-Back Campaign for the At-Risk Segment

- **Finding:** The At-Risk segment comprises 8,711 customers (17.54% of base) who previously spent $140.7 million ($16,152 average spend across ~5.9 orders), but have not bought in an average of 411.5 days (over 13 months).
- **Action:** Alfido Tech should launch a personalized, multi-stage re-engagement email sequence offering an exclusive 15% return incentive and highlighting trending items in their preferred categories (Clothing/Books).
- **Expected Benefit:** Reactivating 10% to 15% of these high-frequency buyers recaptures between $14M and $21M in lifetime transaction volume and halts permanent attrition.

Recommendation 2 – VIP Loyalty Program for Champions to Maximize Advocacy

- **Finding:** Champions represent 8,739 customers (17.59% of base) but generate $190.4 million (27.94% of total revenue) with the highest average spend ($21,785) and an average recency of just 61.3 days.
- **Action:** Implement an exclusive tiered VIP program offering early product drops, priority customer support, zero-fee expedited shipping, and referral rewards.
- **Expected Benefit:** Deepens brand loyalty, increases purchase frequency from 7.6 to 8.5+ orders annually, and converts top spenders into organic brand advocates.

Recommendation 3 – Upselling and Nurturing Potential / Recent Buyers

- **Finding:** The Potential / Recent Buyers segment includes 8,754 customers (17.62% of base) who purchased recently (average recency 114.2 days) but have lower lifetime frequency (3.20 orders) and spend ($8,719 avg).
- **Action:** Deliver personalized post-purchase follow-up recommendations, cross-category bundle discounts (e.g. buy Clothing, get 20% off Books), and time-limited next-order vouchers within 30 days of purchase.
- **Expected Benefit:** Migrates high-recency occasional buyers into the Loyal Customers tier, increasing their average order count from 3.2 towards the 5.8 loyal customer benchmark.

Recommendation 4 – Category Bundling and High-Return Rate Remediation

- **Finding:** Books and Clothing drive 60.0% of total revenue ($409.5M combined across 149,964 orders), yet recorded product returns are notably high at ~49.8% across orders with return records.
- **Action:** Alfido Tech should cross-promote Electronics and Home accessories with top-selling Books and Clothing items, while simultaneously improving sizing guides, material specifications, and customer reviews to address the ~50% return rate.
- **Expected Benefit:** Boosts basket sizes across complementary product lines while reducing return logistics costs and customer friction.

Recommendation 5 – Automated Lifecycle Monitoring and Recency-Triggered Alerts

- **Finding:** Across the dataset, 20.0% of customers (9,942 accounts) are flagged as churned, and inactive customers average 564.3 days since last interaction, indicating customer disengagement often goes unaddressed for over a year.
- **Action:** Build an automated CRM dashboard tracking customer recency thresholds, triggering dynamic reminder notifications when a customer exceeds 180 days without a transaction.
- **Expected Benefit:** Intervenes before customers transition from active/loyal status into dormant or lost states, reducing the overall 20% churn rate by an estimated 3–5 percentage points.

7. Conclusion & Limitations

Conclusion: This customer behavior analysis demonstrates that Alfido Tech possesses a highly active core base (Champions and Loyal Customers driving 55.8% of revenue). Strategic attention must now shift towards re-engaging the 8,711 At-Risk customers, nurturing Recent Buyers, and addressing high product return rates in core categories.

Project Limitations:

- *No Profit Margin / Cost Data:* The dataset records gross purchase amounts and prices but excludes production, shipping, and marketing cost margins.
- *Return Recording Coverage:* Returns data had 47,596 unrecorded values (imputed as non-returns); actual return costs could differ if unrecorded transactions included unlogged returns.
- *Synthetic / Benchmark Characteristics:* Customer demographics and purchase intervals show even distributions characteristic of simulated benchmark datasets.